

Perceived AI-ness in Advertising: Consumer Perceptions of 2026 Super Bowl Commercials

2026-08-25

Overview

This study examines whether consumers' own subjective impression of the degree to which generative AI is involved in an advertisement's creative production (**perceived AI-ness**) is associated with how they evaluate that advertisement and the advertised brand. Survey data (N=236) was collected using two real advertisement clips from the 2026 Super Bowl (Svedka & Liquid Death), each evaluated by every respondent.

R Set-up

```
library(dplyr)
library(tidyr)
library(psych)
library(sandwich)
library(lmtest)
library(ggplot2)
```

Load Data

The raw dataset contains 236 responses, collected via Qualtrics and cleaned to exclude incomplete responses and those with implausibly short completion times (under 3 minutes).

```
df <- read.csv("data/survey_raw_data.csv", stringsAsFactors = FALSE)
nrow(df)
```

```
## [1] 236
```

Define Scale Levels

Survey responses were exported as text labels rather than numeric values. The vectors below specify the correct response order for each scale used in the questionnaire, used in the next step to convert these text labels into numeric scores.

```
likert_levels <- c("Strongly disagree", "Disagree", "Somewhat disagree", "Neither agree nor disagree", "Somewhat agree", "Agree", "Strongly agree")
fam_levels <- c("Not familiar at all: I don't recognise this brand",
               "Slightly familiar: I recognise the name, but know little else about it",
               "Somewhat familiar: I know the brand and their products, but don't recall purchasing or using their products")
```

```

    "Moderately familiar: I have purchased or consumed their products before",
    "Quite familiar: I have purchased/consumed their products and have also seen their adver
    "Very familiar: I regularly purchase/consume their products and/or actively follow their
    "Extremely familiar: I am a loyal or frequent customer and actively engage with their b

feel_levels <- c("Strongly negative", "Negative", "Somewhat negative", "Neither positive nor negative",
badgood_levels <- c("Extremely bad", "Quite bad", "Slightly bad", "Neither good nor bad", "Slightly good
pleasant_levels <- c("Extremely unpleasant", "Quite unpleasant", "Slightly unpleasant", "Neither pleasant
fav_levels <- c("Extremely unfavourable", "Quite unfavourable", "Slightly unfavourable", "Neither favourable
appeal_levels <- c("Extremely unappealing", "Quite unappealing", "Slightly unappealing", "Neither appealing

```

Recode Text Responses to Numeric

Each text response is converted to its corresponding numeric position (1-7) using the scale definitions above. The binary “seen before” items are also recoded from Yes/No to 1/0 for use as a covariate.

```

df <- df %>%
  mutate(
    across(c(AIN1_SV, AIN2_SV, AIN3_SV, AIN1_LD, AIN2_LD, AIN3_LD,
             CREA1_SV, CREA2_SV, CREA3_SV, CREA1_LD, CREA2_LD, CREA3_LD,
             AUTH1_SV, AUTH2_SV, AUTH3_SV, AUTH1_LD, AUTH2_LD, AUTH3_LD,
             AIATT1, AIATT2, AIATT3, AIATT4, AIATT5),
           ~ as.integer(factor(., levels = likert_levels))),

    SVFAM = as.integer(factor(SVFAM, levels = fam_levels)),
    LDFAM = as.integer(factor(LDFAM, levels = fam_levels)),
    SVFAV = as.integer(factor(SVFAV, levels = feel_levels)),
    LDFAV = as.integer(factor(LDFAV, levels = feel_levels)),

    ADATT1_SV = as.integer(factor(ADATT1_SV, levels = badgood_levels)),
    ADATT1_LD = as.integer(factor(ADATT1_LD, levels = badgood_levels)),
    ADATT2_SV = as.integer(factor(ADATT2_SV, levels = pleasant_levels)),
    ADATT2_LD = as.integer(factor(ADATT2_LD, levels = pleasant_levels)),

    BRATT1_SV = as.integer(factor(BRATT1_SV, levels = fav_levels)),
    BRATT1_LD = as.integer(factor(BRATT1_LD, levels = fav_levels)),
    BRATT2_SV = as.integer(factor(BRATT2_SV, levels = appeal_levels)),
    BRATT2_LD = as.integer(factor(BRATT2_LD, levels = appeal_levels)),

    # Prior exposure (Yes/No -> 1/0)
    SEEN_SV = if_else(SEEN_SV == "Yes", 1L, 0L),
    SEEN_LD = if_else(SEEN_LD == "Yes", 1L, 0L)
  )

```

Reliability Check: Perceived AI-ness

The perceived AI-ness scale included a reverse-scored item (“A human creative team was primarily responsible for producing the visual elements of this advertisement”), which is reverse-coded below and tested

alongside the two other items. This item showed a markedly low corrected item-total correlation, and reliability improved substantially when it was excluded. The final composite therefore uses only the first two items.

```
df <- df %>%
  mutate(AIN3_SV_r = 8 - AIN3_SV,
         AIN3_LD_r = 8 - AIN3_LD)

# 3-item version (reverse-scored item included)
psych::alpha(df %>% select(AIN1_SV, AIN2_SV, AIN3_SV_r))

##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(AIN1_SV, AIN2_SV, AIN3_SV_r))
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean sd median_r
##      0.56      0.61    0.58      0.35 1.6 0.052  5.5 1    0.22
##
##      95% confidence boundaries
##              lower alpha upper
## Feldt      0.45  0.56  0.65
## Duhachek   0.45  0.56  0.66
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## AIN1_SV      0.31      0.33  0.19      0.19 0.48   0.085  NA  0.19
## AIN2_SV      0.33      0.36  0.22      0.22 0.57   0.078  NA  0.22
## AIN3_SV_r    0.76      0.77  0.62      0.62 3.32   0.030  NA  0.62
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean sd
## AIN1_SV 236  0.75  0.82  0.73  0.50  6.1 1.1
## AIN2_SV 236  0.75  0.81  0.71  0.45  5.9 1.3
## AIN3_SV_r 236  0.73  0.63  0.28  0.23  4.4 1.7
##
## Non missing response frequency for each item
##      1 2 3 4 5 6 7 miss
## AIN1_SV 0.00 0.01 0.03 0.04 0.11 0.34 0.46 0
## AIN2_SV 0.00 0.03 0.04 0.05 0.14 0.35 0.40 0
## AIN3_SV_r 0.06 0.10 0.21 0.14 0.18 0.19 0.12 0

psych::alpha(df %>% select(AIN1_LD, AIN2_LD, AIN3_LD_r))

##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(AIN1_LD, AIN2_LD, AIN3_LD_r))
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean sd median_r
##      0.8      0.8    0.77      0.57 4 0.021  3.6 1.4    0.47
##
##      95% confidence boundaries
##              lower alpha upper
## Feldt      0.76  0.8  0.84
```

```
## Duhachek 0.76 0.8 0.85
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## AIN1_LD      0.59      0.60 0.43      0.43 1.5 0.052 NA 0.43
## AIN2_LD      0.63      0.64 0.47      0.47 1.8 0.047 NA 0.47
## AIN3_LD_r     0.89      0.89 0.81      0.81 8.3 0.014 NA 0.81
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean sd
## AIN1_LD 236 0.92 0.90 0.87 0.78 4.0 1.8
## AIN2_LD 236 0.90 0.88 0.85 0.75 4.0 1.8
## AIN3_LD_r 236 0.71 0.75 0.51 0.47 2.9 1.4
##
## Non missing response frequency for each item
##      1 2 3 4 5 6 7 miss
## AIN1_LD 0.08 0.17 0.17 0.11 0.24 0.14 0.08 0
## AIN2_LD 0.08 0.20 0.15 0.14 0.17 0.18 0.08 0
## AIN3_LD_r 0.11 0.34 0.28 0.10 0.10 0.05 0.01 0
```

```
# 2-item version (reverse-scored item excluded) -- final decision
```

```
psych::alpha(df %>% select(AIN1_SV, AIN2_SV))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(AIN1_SV, AIN2_SV))
##
##      raw_alpha std.alpha G6(smc) average_r S/N ase mean sd median_r
##      0.76      0.77      0.62      0.62 3.3 0.03 6 1.1 0.62
##
##      95% confidence boundaries
##      lower alpha upper
## Feldt      0.7 0.76 0.82
## Duhachek    0.7 0.76 0.82
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## AIN1_SV      0.55      0.62 0.39      0.62 1.7 NA 0 0.62
## AIN2_SV      0.71      0.62 0.39      0.62 1.7 NA 0 0.62
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean sd
## AIN1_SV 236 0.89 0.9 0.71 0.62 6.1 1.1
## AIN2_SV 236 0.91 0.9 0.71 0.62 5.9 1.3
##
## Non missing response frequency for each item
##      2 3 4 5 6 7 miss
## AIN1_SV 0.01 0.03 0.04 0.11 0.34 0.46 0
## AIN2_SV 0.03 0.04 0.05 0.14 0.35 0.40 0
```

```
psych::alpha(df %>% select(AIN1_LD, AIN2_LD))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(AIN1_LD, AIN2_LD))
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.89      0.89      0.81      0.81 8.3 0.014    4 1.7      0.81
##
##      95% confidence boundaries
##              lower alpha upper
## Feldt      0.86  0.89  0.92
## Duhachek  0.87  0.89  0.92
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## AIN1_LD      0.79      0.81      0.65      0.81 4.2      NA    0 0.81
## AIN2_LD      0.83      0.81      0.65      0.81 4.2      NA    0 0.81
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## AIN1_LD 236 0.95 0.95 0.85  0.81    4 1.8
## AIN2_LD 236 0.95 0.95 0.85  0.81    4 1.8
##
## Non missing response frequency for each item
##      1  2  3  4  5  6  7 miss
## AIN1_LD 0.08 0.17 0.17 0.11 0.24 0.14 0.08  0
## AIN2_LD 0.08 0.20 0.15 0.14 0.17 0.18 0.08  0
```

Create Composite Variables

Composite scores for each construct were calculated by averaging their constituent items.

```
df <- df %>%
  mutate(
    AIN_SV = (AIN1_SV + AIN2_SV) / 2,
    AIN_LD = (AIN1_LD + AIN2_LD) / 2,
    CREA_SV = (CREA1_SV + CREA2_SV + CREA3_SV) / 3,
    CREA_LD = (CREA1_LD + CREA2_LD + CREA3_LD) / 3,
    AUTH_SV = (AUTH1_SV + AUTH2_SV + AUTH3_SV) / 3,
    AUTH_LD = (AUTH1_LD + AUTH2_LD + AUTH3_LD) / 3,
    ADATT_SV = (ADATT1_SV + ADATT2_SV) / 2,
    ADATT_LD = (ADATT1_LD + ADATT2_LD) / 2,
    BRATT_SV = (BRATT1_SV + BRATT2_SV) / 2,
    BRATT_LD = (BRATT1_LD + BRATT2_LD) / 2,
    AIATT = (AIATT1 + AIATT2 + AIATT3 + AIATT4 + AIATT5) / 5
  )
```

Reliability Check (Remaining Composites)

Reliability for all remaining composite measures was assessed using Cronbach's alpha.

```
psych::alpha(df %>% select(CREA1_SV, CREA2_SV, CREA3_SV))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(CREA1_SV, CREA2_SV, CREA3_SV))
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.78      0.78    0.75      0.54 3.6 0.025  3.9 1.4      0.58
##
##   95% confidence boundaries
##         lower alpha upper
## Feldt    0.73  0.78  0.83
## Duhachek  0.73  0.78  0.83
##
## Reliability if an item is dropped:
##   raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## CREA1_SV    0.73      0.73    0.58      0.58 2.7   0.035   NA  0.58
## CREA2_SV    0.53      0.53    0.36      0.36 1.1   0.061   NA  0.36
## CREA3_SV    0.82      0.82    0.69      0.69 4.5   0.023   NA  0.69
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## CREA1_SV 236  0.82  0.82  0.71   0.60  4.1 1.7
## CREA2_SV 236  0.91  0.91  0.87   0.77  3.6 1.8
## CREA3_SV 236  0.77  0.77  0.59   0.51  4.0 1.7
##
## Non missing response frequency for each item
##      1  2  3  4  5  6  7 miss
## CREA1_SV 0.08 0.14 0.16 0.13 0.25 0.17 0.07  0
## CREA2_SV 0.14 0.21 0.19 0.11 0.19 0.12 0.04  0
## CREA3_SV 0.08 0.14 0.17 0.23 0.18 0.12 0.08  0
```

```
psych::alpha(df %>% select(CREA1_LD, CREA2_LD, CREA3_LD))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(CREA1_LD, CREA2_LD, CREA3_LD))
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.86      0.86    0.81      0.67 6.1 0.016  5.1 1.3      0.68
##
##   95% confidence boundaries
##         lower alpha upper
## Feldt    0.82  0.86  0.88
## Duhachek  0.82  0.86  0.89
##
## Reliability if an item is dropped:
##   raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## CREA1_LD    0.81      0.81    0.68      0.68 4.3   0.025   NA  0.68
## CREA2_LD    0.73      0.74    0.59      0.59 2.8   0.034   NA  0.59
## CREA3_LD    0.85      0.85    0.74      0.74 5.6   0.020   NA  0.74
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## CREA1_LD 236  0.86  0.88  0.79   0.72  5.3 1.3
## CREA2_LD 236  0.91  0.91  0.86   0.79  5.0 1.5
```

```
## CREA3_LD 236 0.87 0.86 0.73 0.68 5.1 1.5
##
## Non missing response frequency for each item
##      1      2      3      4      5      6      7 miss
## CREA1_LD 0.00 0.04 0.08 0.08 0.33 0.31 0.17 0
## CREA2_LD 0.01 0.08 0.07 0.14 0.25 0.32 0.13 0
## CREA3_LD 0.03 0.05 0.08 0.11 0.25 0.30 0.18 0
```

```
psych::alpha(df %>% select(AUTH1_SV, AUTH2_SV, AUTH3_SV))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(AUTH1_SV, AUTH2_SV, AUTH3_SV))
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.9        0.9      0.85      0.75 8.8 0.011  3.1 1.5      0.75
##
##      95% confidence boundaries
##      lower alpha upper
## Feldt      0.87    0.9 0.92
## Duhachek   0.88    0.9 0.92
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## AUTH1_SV      0.85      0.85 0.75      0.75 5.9 0.019  NA 0.75
## AUTH2_SV      0.85      0.85 0.74      0.74 5.7 0.019  NA 0.74
## AUTH3_SV      0.86      0.86 0.75      0.75 6.1 0.018  NA 0.75
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## AUTH1_SV 236 0.91 0.91 0.84 0.80 3.4 1.6
## AUTH2_SV 236 0.92 0.91 0.85 0.80 3.3 1.7
## AUTH3_SV 236 0.91 0.91 0.84 0.79 2.8 1.7
##
## Non missing response frequency for each item
##      1      2      3      4      5      6      7 miss
## AUTH1_SV 0.13 0.22 0.18 0.19 0.18 0.07 0.03 0
## AUTH2_SV 0.15 0.28 0.16 0.14 0.16 0.08 0.03 0
## AUTH3_SV 0.27 0.27 0.17 0.10 0.11 0.06 0.03 0
```

```
psych::alpha(df %>% select(AUTH1_LD, AUTH2_LD, AUTH3_LD))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(AUTH1_LD, AUTH2_LD, AUTH3_LD))
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.89        0.9      0.85      0.74 8.7 0.012  4.4 1.5      0.74
##
##      95% confidence boundaries
##      lower alpha upper
## Feldt      0.87    0.89 0.92
## Duhachek   0.87    0.89 0.92
```

```
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## AUTH1_LD      0.83      0.83   0.72      0.72 5.0   0.022   NA  0.72
## AUTH2_LD      0.85      0.85   0.74      0.74 5.7   0.020   NA  0.74
## AUTH3_LD      0.87      0.87   0.77      0.77 6.8   0.017   NA  0.77
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## AUTH1_LD 236  0.91  0.92  0.87   0.82 4.5 1.5
## AUTH2_LD 236  0.91  0.91  0.84   0.80 4.5 1.7
## AUTH3_LD 236  0.90  0.90  0.82   0.77 4.3 1.8
##
## Non missing response frequency for each item
##      1  2  3  4  5  6  7 miss
## AUTH1_LD 0.02 0.11 0.17 0.15 0.24 0.25 0.06  0
## AUTH2_LD 0.05 0.15 0.08 0.11 0.27 0.23 0.11  0
## AUTH3_LD 0.07 0.13 0.14 0.13 0.22 0.22 0.09  0
```

```
psych::alpha(df %>% select(ADATT1_SV, ADATT2_SV))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(ADATT1_SV, ADATT2_SV))
##
##      raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.96      0.96      0.93      0.93 25 0.005  3.4 1.7      0.93
##
##      95% confidence boundaries
##      lower alpha upper
## Feldt      0.95  0.96  0.97
## Duhachek  0.95  0.96  0.97
##
## Reliability if an item is dropped:
##      raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## ADATT1_SV      0.95      0.93   0.86      0.93 13      NA    0  0.93
## ADATT2_SV      0.90      0.93   0.86      0.93 13      NA    0  0.93
##
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## ADATT1_SV 236  0.98  0.98  0.94   0.93 3.4 1.7
## ADATT2_SV 236  0.98  0.98  0.94   0.93 3.4 1.7
##
## Non missing response frequency for each item
##      1  2  3  4  5  6  7 miss
## ADATT1_SV 0.12 0.27 0.20 0.09 0.16 0.12 0.04  0
## ADATT2_SV 0.13 0.25 0.19 0.14 0.13 0.12 0.03  0
```

```
psych::alpha(df %>% select(ADATT1_LD, ADATT2_LD))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(ADATT1_LD, ADATT2_LD))
```

```
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##     0.93     0.93    0.88     0.88  14 0.0086  4.7 1.5     0.88
##
##   95% confidence boundaries
##         lower alpha upper
## Feldt    0.91  0.93  0.95
## Duhachek 0.92  0.93  0.95
##
## Reliability if an item is dropped:
##       raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## ADATT1_LD    0.81    0.88    0.77    0.88 7.1     NA    0  0.88
## ADATT2_LD    0.94    0.88    0.77    0.88 7.1     NA    0  0.88
##
## Item statistics
##       n raw.r std.r r.cor r.drop mean  sd
## ADATT1_LD 236  0.97  0.97  0.91   0.88  4.8 1.5
## ADATT2_LD 236  0.97  0.97  0.91   0.88  4.5 1.6
##
## Non missing response frequency for each item
##       1  2  3  4  5  6  7 miss
## ADATT1_LD 0.04 0.06 0.11 0.11 0.28 0.33 0.07  0
## ADATT2_LD 0.06 0.06 0.15 0.15 0.24 0.25 0.08  0
```

```
psych::alpha(df %>% select(BRATT1_SV, BRATT2_SV))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(BRATT1_SV, BRATT2_SV))
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##     0.95     0.95    0.91     0.91  19 0.0065  3.4 1.6     0.91
##
##   95% confidence boundaries
##         lower alpha upper
## Feldt    0.93  0.95  0.96
## Duhachek 0.94  0.95  0.96
##
## Reliability if an item is dropped:
##       raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## BRATT1_SV    0.84    0.91    0.82    0.91 9.6     NA    0  0.91
## BRATT2_SV    0.97    0.91    0.82    0.91 9.6     NA    0  0.91
##
## Item statistics
##       n raw.r std.r r.cor r.drop mean  sd
## BRATT1_SV 236  0.97  0.98  0.93   0.91  3.4 1.6
## BRATT2_SV 236  0.98  0.98  0.93   0.91  3.4 1.7
##
## Non missing response frequency for each item
##       1  2  3  4  5  6  7 miss
## BRATT1_SV 0.12 0.24 0.20 0.20 0.11 0.11 0.03  0
## BRATT2_SV 0.15 0.22 0.17 0.19 0.13 0.10 0.05  0
```

```
psych::alpha(df %>% select(BRATT1_LD, BRATT2_LD))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(BRATT1_LD, BRATT2_LD))
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.95      0.95   0.91      0.91  19 0.0065  4.3 1.6      0.91
##
##   95% confidence boundaries
##           lower alpha upper
## Feldt      0.93 0.95 0.96
## Duhachek    0.94 0.95 0.96
##
## Reliability if an item is dropped:
##           raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## BRATT1_LD      0.85      0.91   0.82      0.91 9.5      NA    0 0.91
## BRATT2_LD      0.97      0.91   0.82      0.91 9.5      NA    0 0.91
##
## Item statistics
##           n raw.r std.r r.cor r.drop mean  sd
## BRATT1_LD 236 0.97 0.98 0.93 0.91 4.4 1.6
## BRATT2_LD 236 0.98 0.98 0.93 0.91 4.3 1.7
##
## Non missing response frequency for each item
##           1 2 3 4 5 6 7 miss
## BRATT1_LD 0.07 0.08 0.13 0.19 0.26 0.22 0.05 0
## BRATT2_LD 0.09 0.09 0.12 0.14 0.29 0.19 0.07 0
```

```
psych::alpha(df %>% select(AIATT1, AIATT2, AIATT3, AIATT4, AIATT5))
```

```
##
## Reliability analysis
## Call: psych::alpha(x = df %>% select(AIATT1, AIATT2, AIATT3, AIATT4,
##   AIATT5))
##
##   raw_alpha std.alpha G6(smc) average_r S/N   ase mean  sd median_r
##      0.84      0.83   0.86      0.5 4.9 0.014   5 1.3      0.45
##
##   95% confidence boundaries
##           lower alpha upper
## Feldt      0.81 0.84 0.87
## Duhachek    0.82 0.84 0.87
##
## Reliability if an item is dropped:
##           raw_alpha std.alpha G6(smc) average_r S/N alpha se var.r med.r
## AIATT1      0.88      0.87   0.87      0.63 6.9   0.012 0.047 0.65
## AIATT2      0.84      0.82   0.84      0.53 4.5   0.014 0.108 0.53
## AIATT3      0.75      0.75   0.77      0.42 2.9   0.024 0.046 0.41
## AIATT4      0.78      0.77   0.81      0.46 3.4   0.022 0.051 0.45
## AIATT5      0.76      0.75   0.78      0.43 3.1   0.024 0.050 0.43
##
```

```
## Item statistics
##      n raw.r std.r r.cor r.drop mean  sd
## AIATT1 236  0.47  0.56  0.39  0.32  5.6 1.1
## AIATT2 236  0.67  0.72  0.61  0.53  6.1 1.4
## AIATT3 236  0.92  0.88  0.90  0.85  4.6 1.8
## AIATT4 236  0.87  0.83  0.81  0.76  4.2 2.0
## AIATT5 236  0.91  0.87  0.88  0.82  4.4 1.9
##
## Non missing response frequency for each item
##      1  2  3  4  5  6  7 miss
## AIATT1 0.00 0.02 0.04 0.03 0.29 0.42 0.19  0
## AIATT2 0.03 0.02 0.01 0.02 0.09 0.36 0.46  0
## AIATT3 0.08 0.08 0.10 0.12 0.23 0.26 0.12  0
## AIATT4 0.14 0.11 0.11 0.09 0.19 0.23 0.12  0
## AIATT5 0.13 0.08 0.07 0.20 0.19 0.19 0.14  0
```

Descriptive Statistics

Mean and standard deviation for each composite variable were calculated separately by advertisement.

```
df %>%
  summarise(
    AIN_SV_M = mean(AIN_SV), AIN_SV_SD = sd(AIN_SV),
    AIN_LD_M = mean(AIN_LD), AIN_LD_SD = sd(AIN_LD),
    CREA_SV_M = mean(CREA_SV), CREA_SV_SD = sd(CREA_SV),
    CREA_LD_M = mean(CREA_LD), CREA_LD_SD = sd(CREA_LD),
    AUTH_SV_M = mean(AUTH_SV), AUTH_SV_SD = sd(AUTH_SV),
    AUTH_LD_M = mean(AUTH_LD), AUTH_LD_SD = sd(AUTH_LD),
    ADATT_SV_M = mean(ADATT_SV), ADATT_SV_SD = sd(ADATT_SV),
    ADATT_LD_M = mean(ADATT_LD), ADATT_LD_SD = sd(ADATT_LD),
    BRATT_SV_M = mean(BRATT_SV), BRATT_SV_SD = sd(BRATT_SV),
    BRATT_LD_M = mean(BRATT_LD), BRATT_LD_SD = sd(BRATT_LD)
  )

## AIN_SV_M AIN_SV_SD AIN_LD_M AIN_LD_SD CREA_SV_M CREA_SV_SD CREA_LD_M
## 1 6.023305 1.061155 4 1.691657 3.875706 1.446474 5.127119
## CREA_LD_SD AUTH_SV_M AUTH_SV_SD AUTH_LD_M AUTH_LD_SD ADATT_SV_M ADATT_SV_SD
## 1 1.276757 3.134181 1.500317 4.443503 1.53943 3.404661 1.687381
## ADATT_LD_M ADATT_LD_SD BRATT_SV_M BRATT_SV_SD BRATT_LD_M BRATT_LD_SD
## 1 4.684322 1.51339 3.394068 1.643623 4.330508 1.607456
```

Wide-to-Long Transformation

Since each respondent evaluated both advertisements (Svedka & Liquid Death), the dataset is reshaped from one row per respondent to one row per respondent-ad pair, enabling pooled ad-level analysis (472 observations from 236 respondents).

```
df <- df %>% mutate(RespondentID = row_number())

df_long <- df %>%
  pivot_longer(
    cols = c(AIN_SV, AIN_LD, CREA_SV, CREA_LD, AUTH_SV, AUTH_LD,
```

```

        ADATT_SV, ADATT_LD, BRATT_SV, BRATT_LD, SEEN_SV, SEEN_LD),
  names_to = c(".value", "Brand"),
  names_sep = "_"
)

nrow(df_long)

```

```
## [1] 472
```

```
table(df_long$Brand)
```

```
##
## LD SV
## 236 236
```

Derive Brand-matched Covariates

Prior brand familiarity and prior brand favourability were measured for both Svedka and Liquid Death before the ad exposure. Each row is matched to the familiarity/favourability value corresponding to the brand.

```

df_long <- df_long %>%
  mutate(
    FAM = if_else(Brand == "SV", SVFAM, LDFAM),
    FAV = if_else(Brand == "SV", SVFAV, LDFAV)
  )

```

RQ1: Perceived AI-ness and Ad-level Evaluations

The first research question explores how consumers' perceived AI-ness of advertisements relates to their evaluations of those advertisements.

RQ1-1. How is consumers' perceived AI-ness of advertisements associated with their perceived creativity of those advertisements?

RQ1-2. How is consumers' perceived AI-ness of advertisements associated with their perceived authenticity of those advertisements?

RQ1-3. How is consumers' perceived AI-ness of advertisements associated with their overall attitude toward those advertisements?

These three sub-questions are tested using multiple regression, with perceived ad creativity, perceived ad authenticity, and ad attitude each regressed on perceived AI-ness, while controlling for prior brand familiarity, prior brand favourability, and prior exposure to the specific advertisement. As each respondent contributed two evaluations, standard errors are clustered by respondent to account for non-independence.

```

model_rq1_1 <- lm(CREA ~ AIN + FAM + FAV + SEEN, data = df_long)
coeftest(model_rq1_1, vcov = vcovCL(model_rq1_1, cluster = ~RespondentID))

```

```
##
## t test of coefficients:
##
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.672441   0.337669 16.7988 < 2e-16 ***
## AIN         -0.328072   0.035936 -9.1294 < 2e-16 ***
## FAM          0.144745   0.060761  2.3822 0.01761 *
## FAV          0.063253   0.086907  0.7278 0.46709
## SEEN        -0.481809   0.323650 -1.4887 0.13725
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_rq1_2 <- lm(AUTH ~ AIN + FAM + FAV + SEEN, data = df_long)
coeftest(model_rq1_2, vcov = vcovCL(model_rq1_2, cluster = ~RespondentID))
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.719335   0.383903 12.2931 < 2.2e-16 ***
## AIN         -0.392189   0.042292 -9.2734 < 2.2e-16 ***
## FAM          0.218791   0.068059  3.2147 0.001396 **
## FAV          0.165971   0.100943  1.6442 0.100806
## SEEN        -0.070600   0.359583 -0.1963 0.844430
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_rq1_3 <- lm(ADATT ~ AIN + FAM + FAV + SEEN, data = df_long)
coeftest(model_rq1_3, vcov = vcovCL(model_rq1_3, cluster = ~RespondentID))
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.681310   0.407889 11.4769 < 2.2e-16 ***
## AIN         -0.425878   0.041038 -10.3777 < 2.2e-16 ***
## FAM          0.143381   0.065361  2.1937 0.028752 *
## FAV          0.325830   0.103403  3.1511 0.001731 **
## SEEN        -0.411732   0.328418 -1.2537 0.210584
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_rq1_1)$r.squared
```

```
## [1] 0.1821723
```

```
summary(model_rq1_2)$r.squared
```

```
## [1] 0.2306434
```

```
summary(model_rq1_3)$r.squared
```

```
## [1] 0.2417469
```

Across all three outcomes, higher perceived AI-ness was associated with less favourable evaluations. A one-point increase in perceived AI-ness was associated with a 0.33-point decrease in perceived ad creativity ($b = -0.328$, $SE = 0.036$, $p < .001$), a 0.39-point decrease in perceived ad authenticity ($b = -0.392$, $SE = 0.042$, $p < .001$), and a 0.43-point decrease in ad attitude ($b = -0.426$, $SE = 0.041$, $p < .001$), controlling for covariates above. The models explained 18.2%, 23.1%, and 24.2% of the variance across in creativity, authenticity, and ad attitude respectively. Thus, RQ1-1 to RQ1-3 reveal a **consistent negative association between perceived AI-ness and each dimension of ad-level evaluations**.

Among the covariates, prior brand familiarity was positively associated with all three outcomes ($ps < .05$), suggesting that respondents who were more familiar with a brand tended to evaluate its advertisement more favourably across the three dimensions. The corresponding coefficients ranged from $b = 0.143$ to 0.219 .

Prior brand favourability, on the other hand, was significantly associated only with ad attitude ($b = 0.326$, $p = .002$), and not with creativity ($b = 0.063$, $p = .467$) or authenticity ($b = 0.166$, $p = .101$). This suggests that respondents' pre-existing affective disposition toward a brand was more clearly reflected in their overall evaluation of its advertisement than in their more specific judgments of the advertisement's creative or authentic qualities.

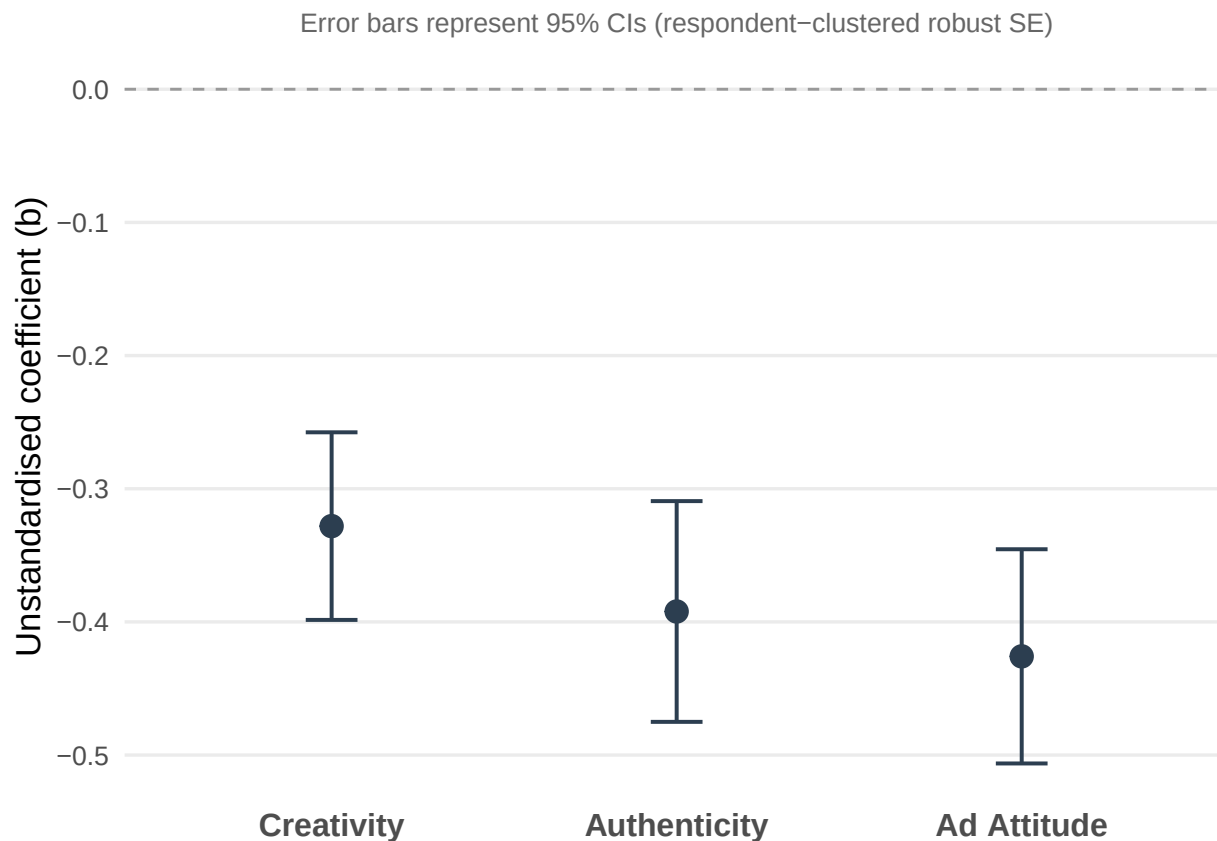
Figure 1: Coefficient Plot

The plot below visualises the association between perceived AI-ness and each ad-level outcome, with error bars representing 95% confidence intervals based on the clustered standard errors above.

```
coef_data <- data.frame(
  Outcome = c("Creativity", "Authenticity", "Ad Attitude"),
  Estimate = c(coef(model_rq1_1)["AIN"], coef(model_rq1_2)["AIN"], coef(model_rq1_3)["AIN"]),
  SE = c(
    coeftest(model_rq1_1, vcov = vcovCL(model_rq1_1, cluster = ~RespondentID))["AIN", "Std. Error"],
    coeftest(model_rq1_2, vcov = vcovCL(model_rq1_2, cluster = ~RespondentID))["AIN", "Std. Error"],
    coeftest(model_rq1_3, vcov = vcovCL(model_rq1_3, cluster = ~RespondentID))["AIN", "Std. Error"]
  )
)

coef_data$Outcome <- factor(coef_data$Outcome, levels = c("Creativity", "Authenticity", "Ad Attitude"))
coef_data$lower <- coef_data$Estimate - 1.96 * coef_data$SE
coef_data$upper <- coef_data$Estimate + 1.96 * coef_data$SE

ggplot(coef_data, aes(x = Outcome, y = Estimate)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey60", linewidth = 0.5) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15, linewidth = 0.7, color = "#2C3E50") +
  geom_point(size = 3.5, color = "#2C3E50") +
  labs(y = "Unstandardised coefficient (b)", x = NULL,
       subtitle = "Error bars represent 95% CIs (respondent-clustered robust SE)") +
  theme_minimal(base_size = 13) +
  theme(plot.subtitle = element_text(size = 10, color = "grey40", hjust = 0.5),
        panel.grid.minor = element_blank(),
        panel.grid.major.x = element_blank(),
        axis.text.x = element_text(size = 12, face = "bold"))
```



Sensitivity Analysis (SV/LD)

To assess whether pooled associations were driven disproportionately by one advertisement, each model was additionally re-estimated separately for Svedka and Liquid Death.

```
# Sensitivity Analysis: SV only
df_sv <- df_long %>% filter(Brand == "SV")

model_sv_crea <- lm(CREA ~ AIN + FAM + FAV + SEEN, data = df_sv)
summary(model_sv_crea)$coefficients["AIN", ]
```

```
##      Estimate   Std. Error    t value    Pr(>|t|)
## -0.263020011  0.085238228  -3.085704803  0.002278487
```

```
model_sv_auth <- lm(AUTH ~ AIN + FAM + FAV + SEEN, data = df_sv)
summary(model_sv_auth)$coefficients["AIN", ]
```

```
##      Estimate   Std. Error    t value    Pr(>|t|)
## -3.481800e-01  8.674527e-02  -4.013822e+00  8.076507e-05
```

```
model_sv_adatt <- lm(ADATT ~ AIN + FAM + FAV + SEEN, data = df_sv)
summary(model_sv_adatt)$coefficients["AIN", ]
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
## -5.197051e-01  9.499463e-02 -5.470890e+00  1.161570e-07
```

```
# Sensitivity Analysis: LD only
```

```
df_ld <- df_long %>% filter(Brand == "LD")
```

```
model_ld_crea <- lm(CREA ~ AIN + FAM + FAV + SEEN, data = df_ld)
summary(model_ld_crea)$coefficients["AIN", ]
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
## -0.186611381  0.048164104 -3.874490850  0.000139212
```

```
model_ld_auth <- lm(AUTH ~ AIN + FAM + FAV + SEEN, data = df_ld)
summary(model_ld_auth)$coefficients["AIN", ]
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
## -2.664611e-01  5.610431e-02 -4.749387e+00  3.582804e-06
```

```
model_ld_adatt <- lm(ADATT ~ AIN + FAM + FAV + SEEN, data = df_ld)
summary(model_ld_adatt)$coefficients["AIN", ]
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
## -2.521231e-01  5.440150e-02 -4.634488e+00  5.982649e-06
```

The negative association between perceived AI-ness and all three outcomes remained statistically significant within each advertisement. For Svedka, perceived AI-ness was associated with lower creativity ($b = -0.263$, $p = .002$), authenticity ($b = -0.348$, $p < .001$), and ad attitude ($b = -0.520$, $p < .001$). For Liquid Death, the corresponding coefficients were -0.187 ($p < .001$), -0.266 ($p < .001$), and -0.252 ($p < .001$), respectively.

The consistent negative direction within both stimuli shows that the pooled RQ1 findings are not solely attributable to overall differences between the two advertisements. The associations were descriptively larger in magnitude for Svedka than for Liquid Death across all three outcomes, particularly for ad attitude ($b = -0.520$ vs. -0.252). However, these between-ad differences in coefficient magnitude were not formally tested and should therefore be interpreted descriptively.

RQ2: Perceived AI-ness and Brand Attitude

The second research question explores how consumers' perceived AI-ness of advertisements relates to their evaluations of the advertised brand(s), and whether this relationship is mediated by perceived ad authenticity.

RQ2-1. How is consumers' perceived AI-ness of advertisements associated with their attitude toward the advertised brands?

RQ2-2. Does perceived ad authenticity mediate the relationship between perceived AI-ness and brand attitude?

The same covariates (prior brand familiarity/favourability, prior ad exposure) are entered consistently across all component models.

```
library(mediation)
```

RQ2-1: Direct Association (Perceived AI-ness -> Brand Attitude)

The direct association between perceived AI-ness and brand attitude is tested using the same pooled, respondent-clustered approach as RQ1.

```
model_rq2_1 <- lm(BRATT ~ AIN + FAM + FAV + SEEN, data = df_long)
coeftest(model_rq2_1, vcov = vcovCL(model_rq2_1, cluster = ~RespondentID))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.540610   0.410834  8.6181 < 2.2e-16 ***
## AIN          -0.353451   0.040569 -8.7124 < 2.2e-16 ***
## FAM           0.101949   0.070272  1.4508  0.1475
## FAV           0.494739   0.097002  5.1003  4.94e-07 ***
## SEEN         -0.229441   0.315399 -0.7275  0.4673
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_rq2_1)$r.squared
```

```
## [1] 0.2175834
```

Higher perceived AI-ness was significantly associated with less favourable brand attitude ($b = -0.353$, $SE = 0.041$, $p < .001$). A one-point increase in perceived AI-ness was associated with a 0.35-point decrease in brand attitude, with the model explaining 21.8% of the variance. Hence, RQ2-1 indicates that the **negative association identified at the advertisement level in RQ1** also extends to consumers' **brand-level evaluations**.

RQ2-2: Mediation Analysis (Perceived AI-ness -> Authenticity -> Brand Attitude)

Two component models are estimated: the mediator model (perceived ad authenticity regressed on perceived AI-ness and covariates) and the outcome model (brand attitude regressed on perceived AI-ness, perceived ad authenticity, and covariates). These are combined using 'mediate()', which computes the indirect effect via 5,000 non-parametric bootstrap resamples with 95% confidence intervals.

'control.value' and 'treat.value' are set to 4 and 5 respectively, rather than the function's default of 0 and 1, since perceived AI-ness is measured on a 1-7 scale and 0 falls outside this range. This anchors the interpretation of all reported effects to a one-point increase in AI-ness within the scale's actual range.

```
set.seed(2026)

model_a_path <- lm(AUTH ~ AIN + FAM + FAV + SEEN, data = df_long)
model_b_path <- lm(BRATT ~ AIN + AUTH + FAM + FAV + SEEN, data = df_long)

mediation_result <- mediation::mediate(
  model.m = model_a_path,
  model.y = model_b_path,
  treat = "AIN",
  mediator = "AUTH",
  control.value = 4,
```

```

    treat.value = 5,
    boot = TRUE,
    sims = 5000
)

summary(mediation_result)

##
## Causal Mediation Analysis
##
## Nonparametric Bootstrap Confidence Intervals with the Percentile Method
##
##           Estimate 95% CI Lower 95% CI Upper p-value
## ACME             -0.2930205   -0.3585061   -0.2321465 <2e-16 ***
## ADE              -0.0604309   -0.1279675    0.0065393  0.0812 .
## Total Effect     -0.3534514   -0.4306917   -0.2746724 <2e-16 ***
## Prop. Mediated    0.8290263    0.6732266    1.0213581 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Sample Size Used: 472
##
##
## Simulations: 5000

```

The key quantities reported are: **ACME** (the indirect effect of perceived AI-ness on brand attitude via authenticity), **ADE** (the remaining direct effect after accounting for authenticity), and **Total Effect** (the sum of the indirect and direct effects).

A significant negative indirect association was observed through perceived ad authenticity (ACME = -0.293, 95% CI [-0.359, -0.232], $p < .001$). After accounting for this indirect pathway, the remaining direct association between perceived AI-ness and brand attitude was substantially reduced and was not statistically significant (ADE = -0.060, 95% CI [-0.128, 0.007], $p = .081$; total effect = -0.353, 95% CI [-0.431, -0.275], $p < .001$). The estimated mediation pathway is visualised in Figure 2. These results suggest that **perceived ad authenticity** accounted for a substantial portion of the observed **negative relationship between perceived AI-ness and brand attitude**, a pattern conventionally described as approaching full mediation.

Sensitivity Check: Mediation by Advertisement

Because each respondent contributes only one row per ad-specific dataset, these models avoid the within-respondent dependence inherent to the pooled mediation model above. To assess whether the pooled mediation pattern was disproportionately driven by either stimulus, the mediation model was re-estimated separately for Svedka and Liquid Death. The same 'control.value' / 'treat.value' anchoring is used for consistency.

```

model_a_sv <- lm(AUTH ~ AIN + FAM + FAV + SEEN, data = df_sv)
model_b_sv <- lm(BRATT ~ AIN + AUTH + FAM + FAV + SEEN, data = df_sv)

# SV Only
mediation_sv <- mediation::mediate(
  model.m = model_a_sv,
  model.y = model_b_sv,
  treat = "AIN",

```

```

mediator = "AUTH",
control.value = 4,
treat.value = 5,
boot = TRUE,
sims = 5000
)
summary(mediation_sv)

##
## Causal Mediation Analysis
##
## Nonparametric Bootstrap Confidence Intervals with the Percentile Method
##
##           Estimate 95% CI Lower 95% CI Upper p-value
## ACME           -0.266089   -0.396101   -0.146044 <2e-16 ***
## ADE            -0.188115   -0.319659   -0.053594  0.0068 **
## Total Effect   -0.454204   -0.624058   -0.280747 <2e-16 ***
## Prop. Mediated  0.585835    0.379157    0.850899 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Sample Size Used: 236
##
##
## Simulations: 5000

```

```

# LD only
model_a_ld <- lm(AUTH ~ AIN + FAM + FAV + SEEN, data = df_ld)
model_b_ld <- lm(BRATT ~ AIN + AUTH + FAM + FAV + SEEN, data = df_ld)

mediation_ld <- mediation::mediate(
  model.m = model_a_ld,
  model.y = model_b_ld,
  treat = "AIN",
  mediator = "AUTH",
  control.value = 4,
  treat.value = 5,
  boot = TRUE,
  sims = 5000
)
summary(mediation_ld)

```

```

##
## Causal Mediation Analysis
##
## Nonparametric Bootstrap Confidence Intervals with the Percentile Method
##
##           Estimate 95% CI Lower 95% CI Upper p-value
## ACME           -0.189922   -0.275494   -0.109400 <2e-16 ***
## ADE            -0.031197   -0.124043    0.057490  0.4832
## Total Effect   -0.221119   -0.341756   -0.107637  0.0004 ***
## Prop. Mediated  0.858915    0.567933    1.440171  0.0004 ***
## ---

```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Sample Size Used: 236
##
##
## Simulations: 5000
```

The indirect association through authenticity remained significant for both Svedka (ACME = -0.266, 95% CI [-0.396, -0.146], $p < .001$) and Liquid Death (ACME = -0.190, 95% CI [-0.275, -0.109], $p < .001$). The presence of a significant indirect pathway within both advertisements provides additional support for the pooled finding, indicating that it was not solely an artefact of combining two stimuli with different levels of perceived AI-ness.

The remaining direct association nevertheless differed descriptively between the two models. For Liquid Death, the ADE was small and non-significant (ADE = -0.031, 95% CI [-0.124, 0.057], $p = .483$), whereas for Svedka a significant negative direct association remained after accounting for authenticity (ADE = -0.188, 95% CI [-0.320, -0.054], $p = .007$). The estimated proportion mediated was correspondingly higher for Liquid Death (0.859) than for Svedka (0.586).

Hence, while authenticity represented a significant pathway in both cases, it accounted for a numerically larger share of the overall association for Liquid Death, whereas perceived AI-ness retained an additional association with brand attitude for Svedka beyond that accounted for by authenticity. This between-ad difference was not formally tested and is therefore interpreted descriptively, rather than as evidence of a moderated mediation effect.

Figure 2: Mediation Path Diagram

To visualise the mediation model, the coefficients for three constituent paths are extracted from the component regression models estimated earlier: the a-path (perceived AI-ness predicting authenticity), the b-path (authenticity predicting brand attitude, controlling for perceived AI-ness), and the c'-path (the remaining direct association between perceived AI-ness and brand attitude, controlling for authenticity).

```
summary(model_a_path)$coefficients["AIN", ]
```

```
##      Estimate   Std. Error    t value    Pr(>|t|)
## -3.921892e-01  4.005558e-02 -9.791126e+00  1.030176e-20
```

```
summary(model_b_path)$coefficients[c("AUTH", "AIN"), ]
```

```
##      Estimate Std. Error  t value    Pr(>|t|)
## AUTH  0.74714070 0.03288846 22.717409 1.862184e-77
## AIN   -0.06043088 0.03125425 -1.933525 5.377687e-02
```

These coefficients ($a = -0.392$, $b = 0.747$, $c' = -0.060$) are used to construct Figure 2, a path diagram illustrating the mediation model reported previously.

```
library(grid)
```

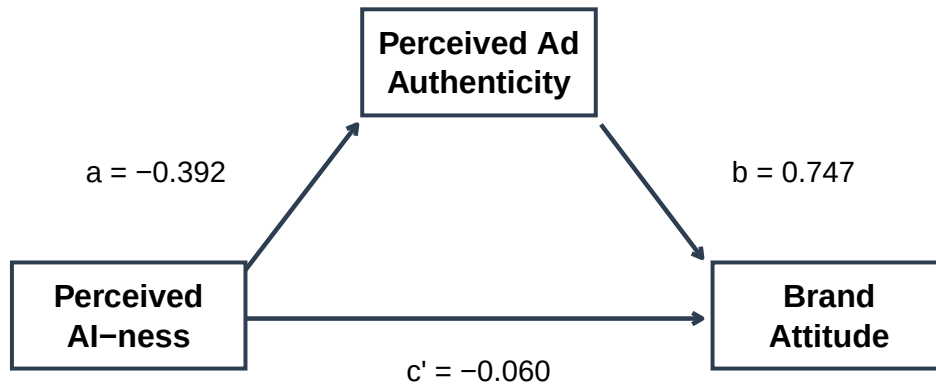
```
ggplot() +
  # Boxes
  annotate("rect", xmin = 1.0, xmax = 2.8, ymin = 2.15, ymax = 2.85, fill = "white", colour = "#2C3E50") +
```

```

annotate("rect", xmin = 3.7, xmax = 5.5, ymin = 3.82, ymax = 4.52, fill = "white", colour = "#2C3E50"
) +
annotate("rect", xmin = 6.4, xmax = 8.2, ymin = 2.15, ymax = 2.85, fill = "white", colour = "#2C3E50"
) +
# Box labels
annotate("text", x = 1.9, y = 2.5, label = "Perceived\nAI-ness", size = 4.1, fontface = "bold", lineheight = 1.2)
) +
annotate("text", x = 4.6, y = 4.17, label = "Perceived Ad\nAuthenticity", size = 4.1, fontface = "bold", lineheight = 1.2)
) +
annotate("text", x = 7.3, y = 2.5, label = "Brand\nAttitude", size = 4.1, fontface = "bold", lineheight = 1.2)
) +
# a path
annotate("segment", x = 2.8, xend = 3.66, y = 2.8, yend = 3.76, colour = "#2C3E50", linewidth = 0.8, lineend = "none")
) +
# b path
annotate("segment", x = 5.54, xend = 6.32, y = 3.76, yend = 2.88, colour = "#2C3E50", linewidth = 0.8, lineend = "none")
) +
# c' path
annotate("segment", x = 2.8, xend = 6.32, y = 2.48, yend = 2.48, colour = "#2C3E50", linewidth = 0.8, lineend = "none")
) +
# Path labels
annotate("text", x = 2.65, y = 3.45, label = "a = -0.392", size = 4.0, hjust = 1)
) +
annotate("text", x = 6.55, y = 3.45, label = "b = 0.747", size = 4.0, hjust = 0)
) +
annotate("text", x = 4.6, y = 2.14, label = "c' = -0.060", size = 4.0)
) +
# Indirect effect
annotate("text", x = 4.6, y = 1.15, label = "Indirect effect (ACME) = -0.293", size = 4.1, fontface = "bold")
) +
annotate("text", x = 4.6, y = 0.78, label = "95% CI [-0.359, -0.232], p < .001", size = 3.8, colour = "#2C3E50")
) +
coord_cartesian(xlim = c(0.6, 8.6), ylim = c(0.35, 4.8), clip = "off")
) +
labs(subtitle = "Indirect effect estimated using 5,000 bootstrap simulations")
) +
theme_minimal(base_size = 13) +
theme(plot.subtitle = element_text(size = 10, colour = "grey40", hjust = 0.5),
      panel.grid = element_blank(),
      axis.text = element_blank(),
      axis.title = element_blank(),
      axis.ticks = element_blank(),
      plot.margin = margin(15, 20, 15, 20))
)

```

Indirect effect estimated using 5,000 bootstrap simulations



Indirect effect (ACME) = -0.293

95% CI [-0.359, -0.232], $p < .001$

RQ3: The Moderating Role of AI Attitude

The final research question examines whether consumers' general attitude toward generative AI moderates the association between their perceived AI-ness of advertisements and each of the ad/brand-level evaluations listed below.

RQ3-1. Do consumers' AI attitudes moderate the relationship between perceived AI-ness and perceived ad creativity?

RQ3-2. Do consumers' AI attitudes moderate the relationship between perceived AI-ness and perceived ad authenticity?

RQ3-3. Do consumers' AI attitudes moderate the relationship between perceived AI-ness and brand attitude?

Each of these questions is tested as an independent simple moderation model via interaction regression. Both perceived AI-ness and AI attitude are mean-centred prior to constructing the interaction terms. This ensures that the main effect of each predictor is interpretable at the mean of the other variable, without altering the interaction coefficient itself or its significance.

```
df_long <- df_long %>%  
  mutate(  
    AIN_c = AIN - mean(AIN, na.rm = TRUE),  
    AIATT_c = AIATT - mean(AIATT, na.rm = TRUE)  
  )
```

RQ3-1: AI Attitude x Perceived AI-ness -> Perceived Ad Creativity

```
model_rq3_1 <- lm(CREA ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_long)
coefTest(model_rq3_1, vcov = vcovCL(model_rq3_1, cluster = ~RespondentID))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error  t value  Pr(>|t|)
## (Intercept)   4.102219   0.297480  13.7899 < 2.2e-16 ***
## AIN_c         -0.343022   0.032532 -10.5442 < 2.2e-16 ***
## AIATT_c        0.335397   0.050141   6.6890 6.467e-11 ***
## FAM           0.095343   0.054660   1.7443  0.08177 .
## FAV           0.059836   0.079515   0.7525  0.45212
## SEEN          -0.266543   0.277972  -0.9589  0.33812
## AIN_c:AIATT_c  0.113306   0.022026   5.1442 3.968e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_rq3_1)$r.squared
```

```
## [1] 0.2995532
```

For perceived ad creativity, the interaction between perceived AI-ness and AI attitude was positive and statistically significant ($b = 0.113$, $SE = 0.022$, $p < .001$), with the model explaining 30.0% of the variance in creativity evaluations.

RQ3-2: AI Attitude x Perceived AI-ness -> Perceived Ad Authenticity

```
model_rq3_2 <- lm(AUTH ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_long)
coefTest(model_rq3_2, vcov = vcovCL(model_rq3_2, cluster = ~RespondentID))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error  t value  Pr(>|t|)
## (Intercept)   2.889497   0.314590   9.1850 < 2.2e-16 ***
## AIN_c         -0.413414   0.037861 -10.9194 < 2.2e-16 ***
## AIATT_c        0.449055   0.052819   8.5017 2.565e-16 ***
## FAM           0.158982   0.058553   2.7152  0.006871 **
## FAV           0.152519   0.087076   1.7516  0.080507 .
## SEEN          0.139023   0.323947   0.4292  0.668010
## AIN_c:AIATT_c  0.083163   0.025173   3.3037  0.001028 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_rq3_2)$r.squared
```

```
## [1] 0.3703587
```

The interaction between perceived AI-ness and AI attitude was positive and statistically significant ($b = 0.083$, $SE = 0.025$, $p = .001$), with the model explaining 37.0% of variance in authenticity evaluations.

RQ3-3: AI Attitude x Perceived AI-ness -> Brand Attitude

```
model_rq3_3 <- lm(BRATT ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_long)
coeftest(model_rq3_3, vcov = vcovCL(model_rq3_3, cluster = ~RespondentID))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.868818   0.330625   5.6524 2.764e-08 ***
## AIN_c         -0.372351   0.037227 -10.0023 < 2.2e-16 ***
## AIATT_c        0.418614   0.048574   8.6181 < 2.2e-16 ***
## FAM           0.041547   0.058127   0.7148  0.4751
## FAV           0.488714   0.088137   5.5449 4.938e-08 ***
## SEEN          0.023642   0.267594   0.0884  0.9296
## AIN_c:AIATT_c  0.127820   0.022070   5.7915 1.286e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_rq3_3)$r.squared
```

```
## [1] 0.3543701
```

The interaction between perceived AI-ness and AI attitude was positive and statistically significant ($b = 0.128$, $SE = 0.022$, $p < .001$), with the model explaining 35.4% of variance in brand attitude.

The key term of interest in each model is the interaction coefficient ('AIN_c:AIATT_c'). A significant positive interaction indicates that a more favourable AI attitude weakens (buffers) the negative association between perceived AI-ness and the outcome; a significant negative interaction would indicate it strengthens that negative association.

Sensitivity Analysis (SV/LD)

Each interaction model is additionally re-estimated separately for each brand. As each respondent contributes only one row per ad-specific dataset, clustered standard errors are not required here. It should be noted that these ad-specific models have substantially less statistical power than the pooled models ($n = 236$ vs. $N = 472$), and interaction effects are typically harder to detect than main effects; a weaker or non-significant result in a single ad subset does not, on its own, contradict a significant pooled finding.

```
df_sv <- df_sv %>%
  mutate(
    AIN_c = AIN - mean(AIN, na.rm = TRUE),
    AIATT_c = AIATT - mean(AIATT, na.rm = TRUE)
  )
df_ld <- df_ld %>%
  mutate(
    AIN_c = AIN - mean(AIN, na.rm = TRUE),
    AIATT_c = AIATT - mean(AIATT, na.rm = TRUE)
  )

# SV only
summary(lm(CREA ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_sv))$coefficients["AIN_c:AIATT_c", ]
```

```
##      Estimate Std. Error      t value      Pr(>|t|)
## 0.167042142 0.062933572 2.654261274 0.008505558
```

```
summary(lm(AUTH ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_sv))$coefficients["AIN_c:AIATT_c", ]
```

```
##      Estimate Std. Error      t value      Pr(>|t|)
## 0.11065294 0.06274609 1.76350328 0.07914920
```

```
summary(lm(BRATT ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_sv))$coefficients["AIN_c:AIATT_c", ]
```

```
##      Estimate Std. Error      t value      Pr(>|t|)
## 0.12426697 0.06554586 1.89587819 0.05923521
```

```
# LD only
```

```
summary(lm(CREA ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_ld))$coefficients["AIN_c:AIATT_c", ]
```

```
##      Estimate Std. Error      t value      Pr(>|t|)
## 1.533318e-01 3.330131e-02 4.604378e+00 6.860621e-06
```

```
summary(lm(AUTH ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_ld))$coefficients["AIN_c:AIATT_c", ]
```

```
##      Estimate Std. Error      t value      Pr(>|t|)
## 0.1358996956 0.0380286087 3.5736173419 0.0004292482
```

```
summary(lm(BRATT ~ AIN_c * AIATT_c + FAM + FAV + SEEN, data = df_ld))$coefficients["AIN_c:AIATT_c", ]
```

```
##      Estimate Std. Error      t value      Pr(>|t|)
## 1.950127e-01 3.930961e-02 4.960941e+00 1.367659e-06
```

The interaction coefficients were positive across all six ad-specific models, consistent with the direction observed in the pooled analyses. For Liquid Death, the interaction was statistically significant for creativity ($b = 0.153$, $p < .001$), authenticity ($b = 0.136$, $p < .001$), and brand attitude ($b = 0.195$, $p < .001$). For Svedka, the interaction was significant for creativity ($b = 0.167$, $p = .009$) yet did not reach the conventional .05 threshold for authenticity ($b = 0.111$, $p = .079$) or brand attitude ($b = 0.124$, $p = .059$).

Visualisation: Interaction Plots (Simple Slopes by AI Attitude)

To visualise the significant moderation effects observed above, predicted values of each outcome are plotted against perceived AI-ness at three representative levels of AI attitude: one standard deviation below the mean (Low), at the mean (Moderate), and one standard deviation above the mean (High).

```
mean(df_long$AIATT, na.rm = TRUE)
```

```
## [1] 4.981356
```

```
sd(df_long$AIATT, na.rm = TRUE)
```

```
## [1] 1.311452
```

```

aiatt_mean <- mean(df_long$AIATT, na.rm = TRUE)
aiatt_sd <- sd(df_long$AIATT, na.rm = TRUE)
ain_mean <- mean(df_long$AIN, na.rm = TRUE)

aiatt_levels <- c(
  Low = aiatt_mean - aiatt_sd,
  Moderate = aiatt_mean,
  High = aiatt_mean + aiatt_sd
)

fam_mean <- mean(df_long$FAM, na.rm = TRUE)
fav_mean <- mean(df_long$FAV, na.rm = TRUE)
seen_mean <- mean(df_long$SEEN, na.rm = TRUE)
ain_range <- range(df_long$AIN, na.rm = TRUE)

```

A helper function generates predicted values across the observed range of perceived AI-ness, holding covariates at their sample means, for each of the three AI attitude levels above. Perceived AI-ness is re-centred here using the same sample mean used to fit the regression models, ensuring the predicted values are consistent with the fitted coefficients.

```

make_interaction_data <- function(model, aiatt_levels, ain_range, fam_mean, fav_mean, seen_mean) {
  do.call(rbind, lapply(names(aiatt_levels), function(lvl) {
    aiatt_val <- aiatt_levels[[lvl]]
    ain_seq <- seq(min(ain_range), max(ain_range), length.out = 50)
    newdata <- data.frame(
      AIN_c = ain_seq - ain_mean,
      AIATT_c = aiatt_val - aiatt_mean,
      FAM = fam_mean,
      FAV = fav_mean,
      SEEN = seen_mean
    )
    newdata$predicted <- predict(model, newdata = newdata)
    newdata$AIN <- ain_seq
    newdata$AI_Attitude <- factor(lvl, levels = c("Low", "Moderate", "High"))
    newdata
  })))
}

```

Figure 3: Perceived AI-ness x AI Attitude on Perceived Ad Creativity (RQ3-1)

```

plot_data_crea <- make_interaction_data(model_rq3_1, aiatt_levels, ain_range, fam_mean, fav_mean, seen_mean)

ggplot(plot_data_crea, aes(x = AIN, y = predicted, linetype = AI_Attitude)) +
  geom_line(linewidth = 0.9, color = "#2C3E50") +
  labs(x = "Perceived AI-ness", y = "Predicted Perceived Ad Creativity",
       linetype = "AI Attitude",
       subtitle = "Predicted values at low (-1 SD), moderate (mean), and high (+1 SD) AI attitude") +
  theme_minimal(base_size = 13) +
  theme(plot.subtitle = element_text(size = 9, color = "grey40"),
        legend.position = "bottom",
        panel.grid.minor = element_blank())

```

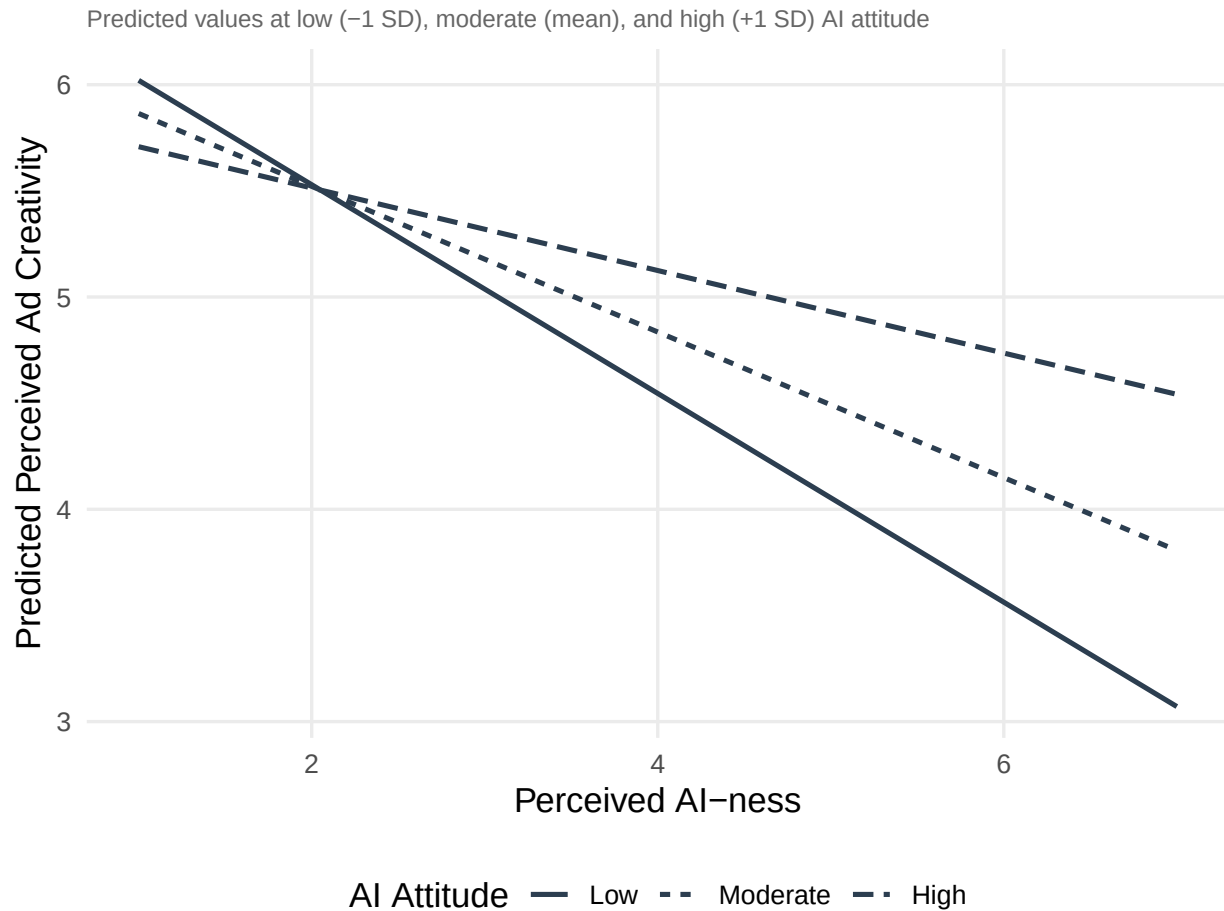


Figure 4: Perceived AI-ness x AI Attitude on Perceived Ad Authenticity (RQ3-2)

```
plot_data_auth <- make_interaction_data(model_rq3_2, aiatt_levels, ain_range, fam_mean, fav_mean, seen_r)

ggplot(plot_data_auth, aes(x = AIN, y = predicted, linetype = AI_Attitude)) +
  geom_line(linewidth = 0.9, color = "#2C3E50") +
  labs(x = "Perceived AI-ness", y = "Predicted Perceived Ad Authenticity",
       linetype = "AI Attitude",
       subtitle = "Predicted values at low (-1 SD), moderate (mean), and high (+1 SD) AI attitude") +
  theme_minimal(base_size = 13) +
  theme(plot.subtitle = element_text(size = 9, color = "grey40"),
        legend.position = "bottom",
        panel.grid.minor = element_blank())
```

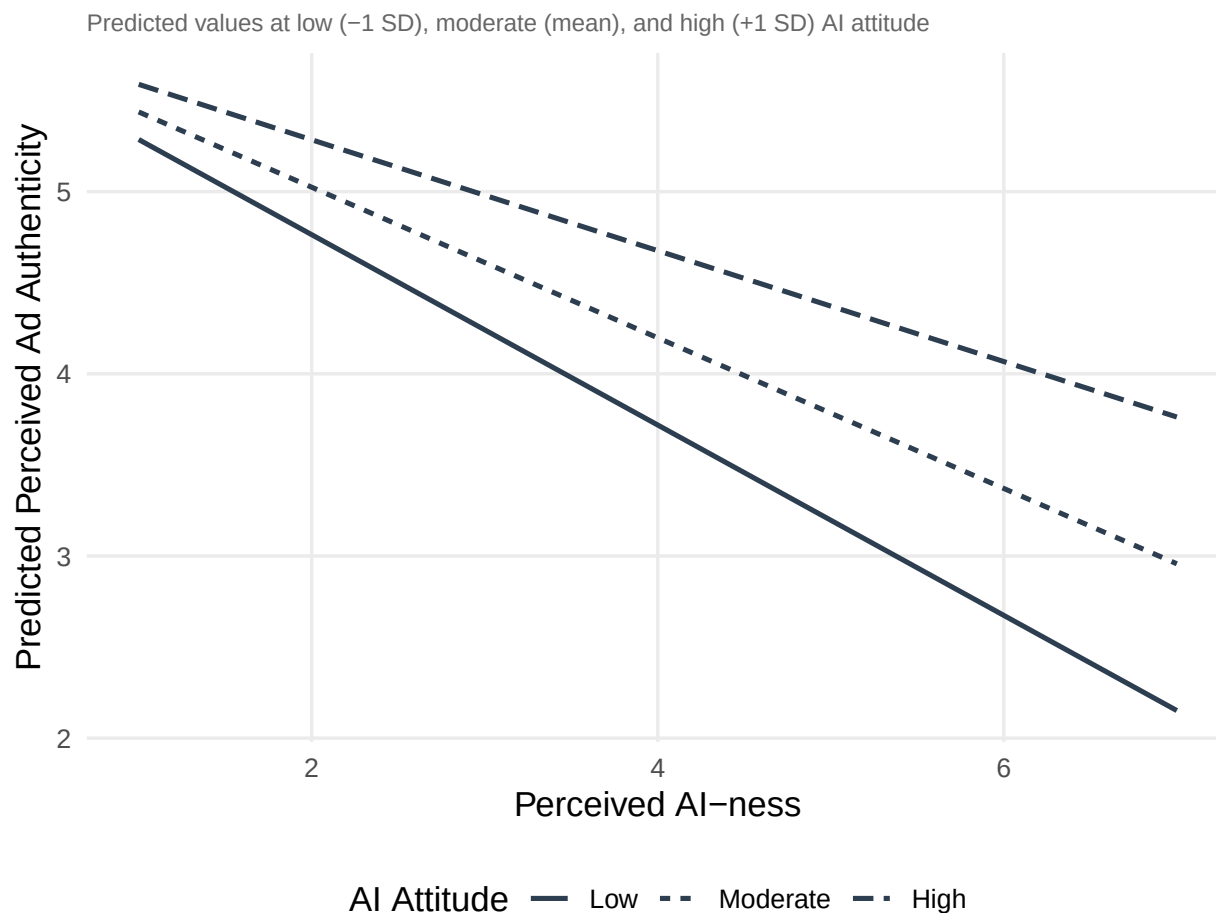
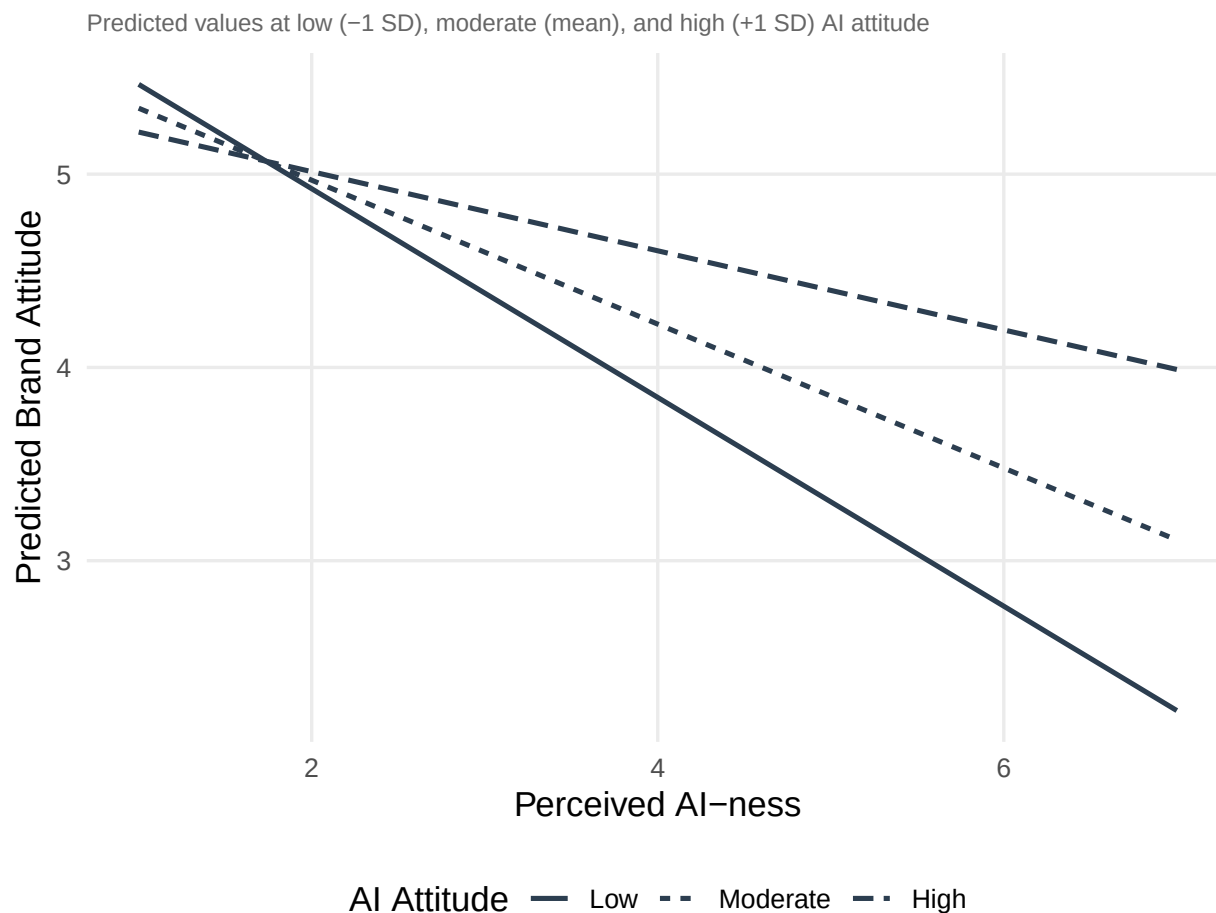


Figure 5: Perceived AI-ness x AI Attitude on Brand Attitude (RQ3-3)

```
plot_data_bratt <- make_interaction_data(model_rq3_3, aiatt_levels, ain_range, fam_mean, fav_mean, seen)

ggplot(plot_data_bratt, aes(x = AIN, y = predicted, linetype = AI_Attitude)) +
  geom_line(linewidth = 0.9, color = "#2C3E50") +
  labs(x = "Perceived AI-ness", y = "Predicted Brand Attitude",
       linetype = "AI Attitude",
       subtitle = "Predicted values at low (-1 SD), moderate (mean), and high (+1 SD) AI attitude") +
  theme_minimal(base_size = 13) +
  theme(plot.subtitle = element_text(size = 9, color = "grey40"),
        legend.position = "bottom",
        panel.grid.minor = element_blank())
```



Across all three figures, the lines diverge as perceived AI-ness increases; the slope is the steepest (most negative) for respondents with low AI attitude, and flattest for those with high AI attitude, consistent with a buffering pattern in which a favourable AI attitude attenuates the negative associations between perceived AI-ness and each outcome.

Simple Slope Estimates: Association between Perceived AI-ness and each outcome at Low/Moderate/High AI Attitude

To complement the interaction plots above, the simple slope of perceived AI-ness is calculated at each of the three AI attitude levels. The simple slope equals the main effect of perceived AI-ness plus the interaction coefficient multiplied by the (centred) AI attitude value.

```
compute_simple_slopes <- function(model, aiatt_levels, aiatt_mean) {
  b_ain <- coef(model)["AIN_c"]
  b_interaction <- coef(model)["AIN_c:AIATT_c"]

  sapply(aiatt_levels, function(level) {
    aiatt_c <- level - aiatt_mean
    b_ain + b_interaction * aiatt_c
  })
}
```

```
# Creativity (RQ3-1)
compute_simple_slopes(model_rq3_1, aiatt_levels, aiatt_mean)
```

```
##      Low.AIN_c Moderate.AIN_c   High.AIN_c
##      -0.4916182   -0.3430222   -0.1944262
```

```
# Authenticity (RQ3-2)
compute_simple_slopes(model_rq3_2, aiatt_levels, aiatt_mean)
```

```
##      Low.AIN_c Moderate.AIN_c   High.AIN_c
##      -0.5224776   -0.4134139   -0.3043502
```

```
# Brand Attitude (RQ3-3)
compute_simple_slopes(model_rq3_3, aiatt_levels, aiatt_mean)
```

```
##      Low.AIN_c Moderate.AIN_c   High.AIN_c
##      -0.5399803   -0.3723510   -0.2047217
```

The simple slopes remain negative across low, mean and high levels of AI attitude, but become progressively less negative as AI attitude increases. This pattern is consistent across creativity (-0.492 to -0.194), authenticity (-0.522 to -0.304), and brand attitude (-0.540 to -0.205), and descriptively supports the interpretation that favourable AI attitudes attenuate the negative association between perceived AI-ness and each outcome. These calculations are descriptive only; no separate simple-slope significance tests are conducted.